

Deepak Mallampati

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Kent, OH - US-based, F-1 OPT, open to SF or Seattle relocation (3d/wk hybrid)

PROFESSIONAL SUMMARY

May 2025 Master's grad (Kent State, Computer Science) shipping production AI features at Progress Solutions: agentic LLM workflows (~25% stability gain), RAG (~35% retrieval precision improvement), predictive ML pipelines (15-20% recall on high-risk patient categories), and FastAPI/Docker services with structured logging (~30% post-deploy defect reduction). Eight shipped/ongoing AI projects spanning consumer-product full-stack and AI-powered features: Manga Lens (Chrome Web Store, multi-LLM-provider abstraction across Claude Sonnet + GPT-4o mini + GPT-4.1 Nano + Ollama, 29 sites), Dream Decoder (full-stack FastAPI + React/TS multimodal app with intermediate structured-prompt transformation lifting alignment ~30% and first-pass success ~25-30%), Agentic Healthcare Claims (5-stage agent pipeline with schema-validated JSON handoffs preventing cascading hallucinations). Strong CS fundamentals (data structures, algorithms, design patterns, systems thinking) and responsible AI practice (HIPAA-conscious data governance, audit trails, de-identification, structured grounding rules). Production discipline matched to public-scale services.

CORE COMPETENCIES

AI-Powered Feature Delivery • Production LLM Integration (Claude / GPT / Replicate / Ollama) • Responsible AI (Privacy / Grounding / Audit Trails) • Full-Stack Python / TypeScript / React • FastAPI + Docker Production • PyTorch / scikit-learn / XGBoost • CS Fundamentals (DSA + Systems Thinking) • Code Review + Cross-Functional Collaboration • Mentorship + Onboarding Playbook (150+ Students) • F-1 OPT, SF / Seattle Relocation Ready

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI / ML Engineer Trainee

USA - Healthcare Technology, AI Consulting, SaaS Healthcare

- Developed **agentic LLM workflows** for multi-step healthcare queries (eligibility, care navigation, documentation clarification) with structured reasoning, tool discipline, and grounding rules; agent stability up ~25%; hallucination reduction >30%.
- Built production **RAG systems** for clinical knowledge retrieval and healthcare documentation search; recursive semantic chunking and transformer-based embeddings improved **contextual retrieval precision** ~35% and reduced irrelevant retrieval >30%; retrieval-grounded response alignment exceeded 90%.
- Designed and shipped a **five-agent Healthcare Claims pipeline** (Intake, Validation, Consistency, Duplicate Detection, Risk Scoring) with **schema-validated JSON contracts (pydantic)** between agents to prevent cascading hallucinations - audit-ready reasoning traces for compliance review.
- Shipped **predictive ML pipelines** (scikit-learn, XGBoost) for patient no-show prediction and care engagement scoring; recall up 15-20% on high-risk categories with precision held >90% via class weighting, stratified sampling, and threshold calibration.
- Packaged ML / LLM inference as **FastAPI / Flask** RESTful services, containerized with **Docker**, with structured logging and load simulation; cut post-deployment defects ~30%.
- Built preprocessing pipelines (Pandas, NumPy) for EHR extracts, appointment histories, and support ticket logs; raised dataset reliability above 98% and cut downstream model instability ~40%.
- Maintained HIPAA-conscious data governance: de-identification, data-lineage documentation, evaluation audit trails, and stakeholder system-limitation docs.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern)

Hyderabad, India - Oil & Gas, Enterprise ERP, Industrial Automation

- Optimized SQL and **T-SQL stored procedures** powering Synthesis Order-to-Cash; reduced query execution time ~**20%** and data retrieval latency ~**25%**.
- Built **CI/CD pipelines with Jenkins** for schema updates and stored-procedure deploys; cut deployment errors >**30%** and improved release-cycle time ~**35-40%**.
- Designed performance dashboards using SQL Server DMVs and Grafana to track long-running queries, deadlocks, CPU/I/O contention; cut incident recurrence ~**25%**.

Suvidha Foundation

Jan 2022 - Mar 2022

Machine Learning Engineer

Hyderabad, India - Nonprofit Tech, Video Analytics, Applied NLP

- Built a **transformer-based video summarization** system over 5,000+ recorded sessions with hierarchical abstractive summarization and timestamp-anchored clip extraction. Manual review time cut **60-70%**; ~**85%** alignment with human-curated highlights.
- Implemented **document Q&A with RAG** (semantic chunking + embedding retrieval); cut hallucinations ~**30%** and lifted contextual answer accuracy >**85%**.
- Deployed the stack as a Flask-based API with a lightweight web interface for non-technical nonprofit staff.

PROJECTS

Agentic Healthcare Claims Processing & Fraud Risk Intelligence System

Five-stage multi-agent pipeline (Intake, Validation, Consistency, Duplicate, Risk Scoring) with schema-validated JSON contracts between agents preventing cascading hallucinations. RAG-grounded CPT/ICD validation against vector-indexed policy documents. ANN similarity for duplicate detection. Audit-ready explainable risk scoring with reasoning traces.

Python, LangChain, FastAPI, vector retrieval (ANN), pgvector, GPT-4 / Claude, schema contracts (pydantic)

Manga Lens - Multi-Provider AI Vision Chrome Extension (shipped solo, Chrome Web Store)

Real-time AI manga / webtoon translation extension shipped solo to the Chrome Web Store. Multi-provider abstraction across 4 AI vision providers (Claude Sonnet / GPT-4o mini / GPT-4.1 Nano / Ollama) with per-provider payload routing (WebP cloud, JPEG Ollama after CUDA crashes), 7-day translation cache, 29 site selectors via Manifest V3 service workers, narrowed host permissions, privacy policy.

TypeScript, Manifest V3, Service Workers, Claude / GPT-4o mini / GPT-4.1 Nano / Ollama, npm

Dream Decoder - Multimodal Creative Intelligence Platform

Full-stack FastAPI + React/TypeScript/Vite/Tailwind app coordinating multimodal APIs (Perplexity Sonar for interpretation, Replicate image models for 16:9 visuals). Intermediate structured-prompt transformation layer lifted contextual alignment ~30% and first-pass image success ~25-30% over naive direct-prompt orchestration.

FastAPI, React + TypeScript + Vite, Tailwind, Perplexity Sonar, Replicate, Docker

Agentic Pixel Character Synthesis & Animation Engine (ongoing)

Phased generative AI system for identity-consistent, pose-controlled pixel character synthesis with sprite-sheet export. Stable Diffusion fine-tuning + LoRA for identity consistency, ControlNet + OpenPose/MediaPipe for pose conditioning, latent-space consistency for animation smoothness, and an LLM-based agent orchestrator that decomposes high-level prompts into generation tasks.

PyTorch, Hugging Face Diffusers, FastAPI, Docker, Stable Diffusion, LoRA, ControlNet

Driver Drowsiness Detection - Real-Time YOLOv8

Replaced a two-stage CNN pipeline with a unified YOLOv8 detect-and-classify model; reduced inference latency ~30%. Sliding-window confidence aggregation cut blink-driven false positives ~25%; adaptive frame skipping and NMS tuning for stable real-time operation.

YOLOv8, OpenCV, PyTorch, LabelImg

EDUCATION

Master of Science, Computer Science - Kent State University Focus: Applied Machine Learning, NLP, Database Systems.	Aug 2023 - May 2025
Bachelor of Technology, Computer Science - KL University Founder / Lead of E-Farming platform (university project, 80-120 active users phase one).	Jun 2019 - May 2023

CERTIFICATIONS

Deep Learning Specialization - <i>DeepLearning.AI</i> (Coursera)	2024
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SKILLS

Languages: Python, TypeScript, JavaScript, SQL (T-SQL, PostgreSQL, SQLite), C++ (Arduino), PHP
AI / ML: PyTorch, scikit-learn, XGBoost, LangChain, LlamaIndex, Hugging Face Transformers, Diffusers, YOLOv8, OpenCV, ControlNet, Stable Diffusion
LLM / GenAI: RAG, agentic workflows, schema-validated structured outputs, prompt engineering, evaluation pipelines, guardrails, grounding rules, embeddings, vector search (FAISS / pgvector / Pinecone-portable), audit-trace reasoning, multi-agent orchestration, multi-provider AI routing (Claude / GPT-4o / Perplexity / Replicate / Ollama)
Backend: FastAPI, Flask, Node.js (Manifest V3 service workers), REST, Docker, structured logging, load simulation
Frontend: React, Vite, Tailwind, Chrome Extensions (Manifest V3 + content scripts + service workers)
Data: PostgreSQL + pgvector, Pandas, NumPy, ANN similarity, embeddings, SQL data pipelines, T-SQL stored procedures
DevOps: Docker, Jenkins, GitHub Actions, Grafana, AWS/GCP/Azure-portable cloud-ready deployment
Domains: Applied AI, Healthcare AI (HIPAA-conscious), Regulated-Domain LLM Systems, Document Processing + RAG, Multi-Agent Validation, Forward Deploy / Customer-Facing Delivery, Nonprofit ML